Catabolism of phenylalanine and tyrosine yields fumarate that also enters TCA cycle for further catabolism. These intermediates that enter into TCA cycle can act as precursors of gluconeogenesis (Figure 4.9). Hence, the amino acids that can contribute their carbon skeletons for the synthesis of glucose are called as glucogenic amino acids. eg. glycine, alanine, aspartate, glutamate, etc. However, certain amino acids yield acetyl CoA and acetoacetyl CoA that can act as substrates of fatty acid synthesis or give rise to ketone bodies. These amino acids are called as ketogenic amino acids. Eg. Leucine and Lysine. Certain amino acids like phenylalanine, isoleucine and tyrosine are both glucogenic and ketogenic.

Regeneration of amino acids

By reductive amination or by reversal of trans-deamination, the amino acids can be re-synthesized from their respective keto-acids.

4.1.6 Transmethylation

One carbon metabolism

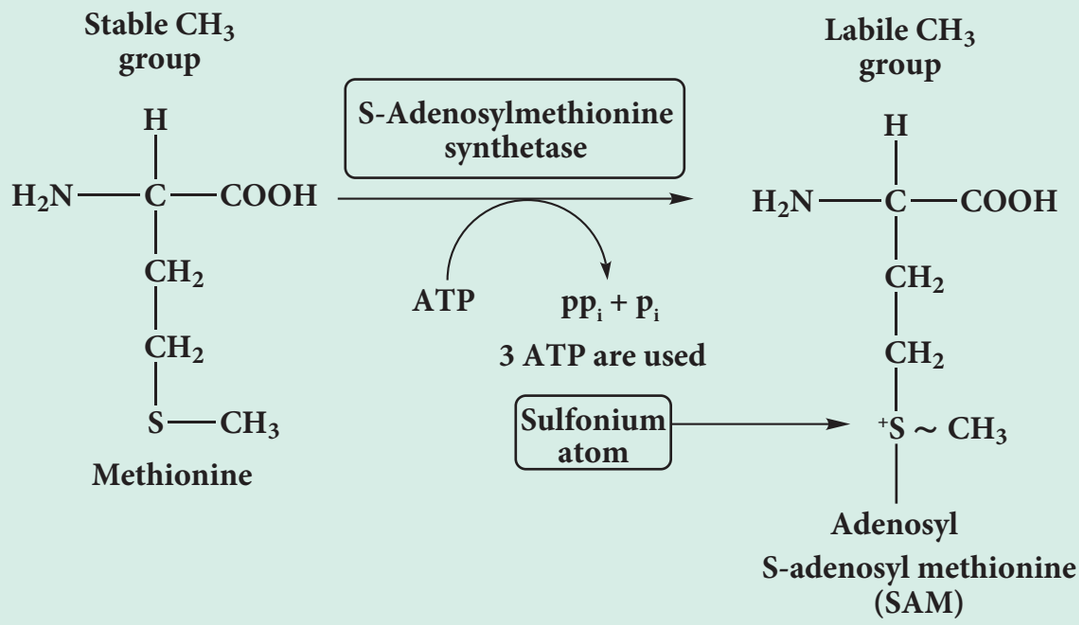

Figure 4.10. Formation of S-adenosyl methionine

One carbon metabolism plays an important role in the synthesis of many compounds like purines and pyrimidines. One carbon pool includes formyl, methyl, methylene, methenyl, formimino and hydroxymethyl groups. Except methyl group, all the other one carbon transfers are mediated by Tetrahydrofolate that bears one carbon moieties at N5 and N10 positions. Formate, tryptophan, glycine, serine and choline are the methyl group donors to tetrahydrofolate. Transfer of methyl group involves S-adenosyl methionine (Figure 4.10), which is formed from methionine

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